

Constructing an Autonomous Navigation System Based on Wind Direction

Emmy Lo, Team 21, Section 2, May 9, 2025

Abstract—This paper discusses the design, testing, and deployment of an autonomous surface vehicle (ASV) built for E80: Experimental Engineering, a course at Harvey Mudd College. An ASV was designed to navigate a body of water using wind direction as its guide. A weather vane sensed the direction that wind was blowing from, and a proportional control feedback loop calculated how much each turning motor needed to spin to orient the ASV in the direction the vane was pointing. The ASV was outfitted with two low-cost sensors, an anemometer and a thermistor, to collect data on its operating conditions and energy consumption. The ASV was deployed in Dana Point, California and gathered data on the performance of the ASV's wind navigation system. The magnitude of steady-state error, the absolute difference between the weather vane output angle and the actual yaw angle of the robot from the inertial measurement unit onboard the ASV, ranged from 1.2 ± 0.4 to 39.6 ± 0.5 degrees. The current, low-cost wind navigation system can turn the ASV in a general direction dictated by the wind, but improvements can be made to achieve higher accuracy.

I. INTRODUCTION

This project was completed as a requirement for E80: Experimental Engineering, a course for second-year undergraduate students at Harvey Mudd College (HMC). The goal was to design an autonomous surface vehicle (ASV) that can orient itself using information about wind direction.

Objects traveling in the same direction that wind is blowing in can be propelled, which requires less energy from the object to move. Traveling opposite this direction causes the object to experience a high amount of drag force. For an ASV, resistance from wind translates into an increase in motor effort and an increased need for power. A system that allows the ASV to orient itself in the same direction as the wind can decrease its energy use and extend the life of its battery.

A system that uses wind to efficiently move a vessel through a body of water has numerous applications. A vehicle that needs to stay out in the water for an extended period of time would benefit from the energy conservation resulting from a decrease in motor use. A probe that monitors weather patterns could find it useful to travel in the same direction as the wind. A vessel could act as a sail-less boat that orients itself to the wind then turns off its motors when the wind is blowing fast enough.

Various low-cost sensors were employed to create a wind-navigation system. A weather vane, an inertial measurement unit (IMU), and a proportional control feedback loop in C++ were used to determine the motor efforts required to orient the ASV in the direction of the wind. Other sensors were used to determine the necessary conditions for operation and the amount of energy consumed by the motors. Wind speed was measured to determine the minimum amount of wind required to rotate the

weather vane and to push the ASV through the water. The temperature of different motor H-bridges were monitored to assess energy consumption by the motors. The data from the sensors will determine if a low-cost wind navigation system is feasible, able to be used in real-world conditions, and energy-saving.

Sensors were tested and calibrated in the HMC E80 laboratory and tank room. The end goal was to deploy the ASV in Dana Point Harbor to assess the performance of the wind-navigation system.

This report will cover the design, testing, and deployment of the ASV. Section II will provide justification for sensor choice and circuit design as well as information about sensor calibration. In Section III, mechanical design and code revisions will be discussed. Section IV will lay out the procedure used for deployment in Dana Point. In Section V, results will be reported and analyzed. Section VI will wrap up the report, draw conclusions, and explore possibilities for future work.

II. SENSOR SELECTION, DESIGN, AND CALIBRATION

A Teensy 4.0 microcontroller was used in this project. The Teensy can read voltages ranging from 0 to 3.3V off its pins and report them in Teensy voltage units [1]. One Teensy voltage unit is equivalent to 0.003237V [2]. The pins are sampled from at a rate of 10Hz [3]. Circuit design was heavily influenced by the Teensy's capabilities.

Additionally, a 5V regulator was used to reduce the 12 Volts from the battery. Where possible, 5V was selected as the supply voltage.

A. Temperature

A temperature sensor was used to measure the heat of the H-bridge on two motors: a rear motor used for turning and a front motor used for driving the ASV forward.

Comparing H-bridge temperatures allows for an assessment of motor energy consumption. H-bridges dissipate power in the form of heat, and their temperature will increase when more current is drawn by the motor [4]. Motors draw more current when the load on them increases, like when moving through water [5]. Thus, an increase in H-bridge temperature suggests an increase in energy consumption.

It is predicted that a turning motor, which is constantly changing speeds, will draw less current and have a lower H-bridge temperature than a driving motor that is always running at the same speed. The energy required to change speeds is expected to be less than the energy required to constantly drive at a high speed. This assessment will help determine if a wind-

based navigation system with motors that are constantly rotating the ASV is truly energy-saving.

The expected range of H-bridge temperatures was 20 to 80°C. According to the H-bridge datasheet, the operating temperature range is -40 to 125 °C [6]. To set the lower limit, it was assumed that the H-bridge, which was inside the waterproof box, would never dip below outside temperatures. The upper limit was chosen to be 80°C to keep the temperature range narrow. Based on the length of time that the motors were run, temperatures higher than 80°C were not expected.

The Murata NXFT15WB473FA2B150 thermistor was selected to measure H-bridge temperature. It is inexpensive, has an accuracy of $\pm 1\%$, and works within the desired range. It has a thermal time constant of 4 seconds, which works for the application because rapid changes in temperature are not expected [7]. Its small size made it easy to adhere to the H-bridge with a piece of electrical tape. Additionally, the output of thermistors, resistance, can be easily converted to temperature using a voltage divider circuit (Fig. 1).

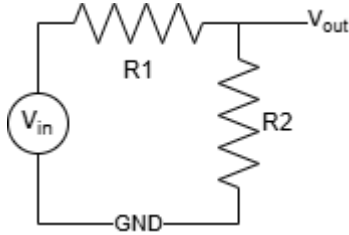


Fig. 1. Voltage Divider Circuit Schematic

To find the range of the voltage divider output, the temperature of the upper and lower limits of the temperature range were converted to thermistor resistance output.

$$R = R_0 \exp\left(B\left(\frac{1}{T} - \frac{1}{T_0}\right)\right) \quad (1)$$

Where R is the current resistance of the thermistor at temperature T in Kelvins, R_0 is the known resistance at reference temperature T_0 in Kelvins, and B is a constant from the datasheet [7].

Equation (1) is a simplified version of the Steinhart-Hart equation [8]. It is used to calculate expected thermistor output at a given temperature. Using (1), the resistances are expected to range from 5.67k Ω to 59.25k Ω . In the voltage divider circuit, the thermistor will be R_1 (Fig. 1). The average of this range, rounded to the nearest whole number, 32k Ω , is the value of R_2 (Fig. 1). An input voltage of 5V will yield a voltage divider output range of 1.75V to 4.25V.

Since these voltages are outside the range that the Teensy is capable of reading, 0 to 3.3V, the output voltage must be shifted and scaled using an offset amplifier. For this circuit, the desired offset is 2.20V and the desired gain is 0.6.

$$\left(1 + \frac{R_f}{R_{n1}}\right)\left(\frac{R_g}{R_{p1} + R_g}\right)V_{p1} - \frac{R_f}{R_{n1}}V_{n1} = V_{out} \quad (2)$$

where R_f , R_{n1} , R_g , and R_{p1} are resistors with positions defined in Fig. 2, V_{p1} is the voltage on the non-inverting side of the op-amp, V_{n1} is the voltage on the inverting side of the op-amp, and V_{out} is the voltage output

The general equation for an offset amplifier (2) was used to set the resistor values in the thermistor circuit (Fig. 2). The voltage output of the circuit was 0.975V to 2.48V, which does not use the whole range of Teensy values. This design choice was made to ensure that voltages from temperatures in the expected range were properly read by the Teensy.

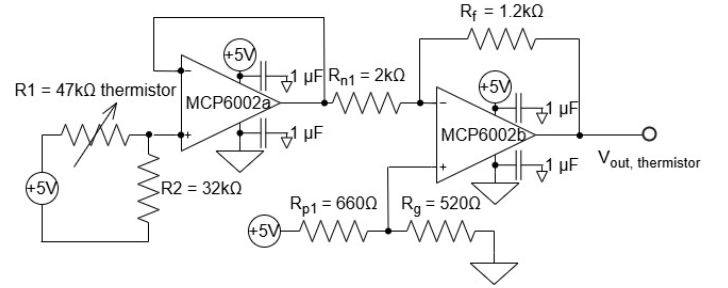


Fig. 2. Thermistor Circuit Schematic

To minimize the effects of loading from the Teensy and motherboard, a unity gain buffer was inserted before the offset amplifier. The MCP6002 was selected for this application because of its low cost and ability to contain two op-amps in one chip [9]. No modifications were made to the code, and taking 10 samples from the circuit per second was sufficient for the measurement. Therefore, the 10Hz sampling rate of the Teensy was appropriate.

To convert voltage output from the Teensy to temperature, a calibration curve was created. The thermistor was placed into an insulated cup of water and allowed to adjust to the temperature for about 15 seconds. The voltage output of the circuit was recorded. This process was repeated for a total of 8 temperatures inside of the 20 to 80°C range. The data was run through a curve-fitting MATLAB script. It was determined that the linear fit was best for this temperature range because the residuals appeared random. Fig. 3 displays the line fitted to the data as well as the calibration equation.

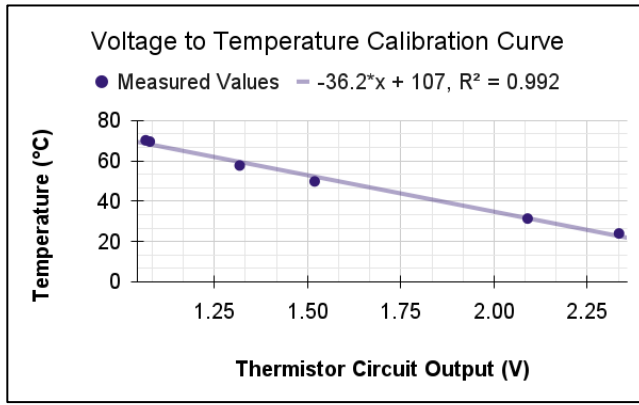


Fig. 3. Thermistor Calibration Curve

B. Wind Direction and ASV Orientation

To construct a navigation system that uses wind direction as an input, wind direction must be measured. A 3D-printed weather vane mounted onto a potentiometer served as the wind direction sensor [10]. The 10kΩ potentiometer chosen is capable of making 10 full rotations. Each counterclockwise rotation adds 1kΩ of resistance. To measure the output resistance of the potentiometer due to the wind, a voltage divider circuit was employed (Fig. 1).

The potentiometer occupied the R1 position. It was set to start at its midpoint, 5kΩ, and this was the value for R2 (Fig. 1). Due to the mechanical stiffness of the potentiometer and the predicted wind speeds at the target deployment site, Dana Point Harbor, the assumption was made that the wind would not be powerful enough to blow the weather vane around more than three full rotations in either the clockwise or counterclockwise direction. Therefore, the expected resistance output is 2kΩ to 8kΩ. For an input voltage of 5V, the potential range of outputs from the divider was 1.92V to 3.57V.

The voltage divider output must be transformed with an offset amplifier to fit into the Teensy range. The optimal gain is 1.83 and the optimal offset is 2.45V. Resistors were selected to produce these values (Fig. 4). The output of the potentiometer circuit is 0.52V to 3.3V.

Like in the thermistor circuit, a unity gain buffer was inserted before the offset amplifier to reduce the effects of loading. Because there are two op-amps in the circuit, the MCP6002 was chosen for this circuit too (Fig. 4).

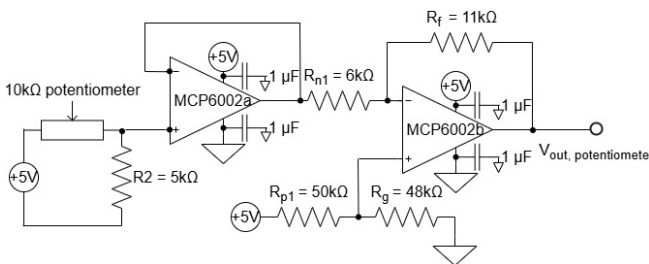


Fig. 4. Weather Vane Circuit Schematic

Since a single turn of the potentiometer is 1kΩ, calculating the angle of the weather vane was a straightforward process. First, the voltage read from the Teensy pin was converted from Teensy units to Volts.

$$\frac{6k\Omega}{11k\Omega} \left[\left(1 + \frac{11k\Omega}{6k\Omega} \right) \left(\frac{48k\Omega}{50k\Omega + 48k\Omega} \right) 5V - V_T \right] = \frac{5k\Omega}{R_{po} + 5k\Omega} 5V \quad (3)$$

where V_T is the voltage measured by the Teensy in Volts and R_{po} is the output resistance of the potentiometer

Then, (3) was used to determine potentiometer resistance from the voltage read off of the Teensy pin. This equation came from combining (2) and the voltage divider output equation and rearranging terms.

The position at each multiple of 1kΩ was considered the starting point, 0 degrees. Counterclockwise rotation is defined as the positive direction. The angle of rotation can be obtained by finding the number of Ohms the potentiometer outputs away from the starting point. For example, if the potentiometer resistance is 3750Ω, then the starting point is 3000Ω. 750Ω ohms is three-quarters of 1kΩ, which is one full turn. Therefore, the vane has rotated three-fourths of 360 degrees, or 270 degrees counterclockwise.

A MATLAB script was written to convert Teensy output to angle in degrees. A similar script was also coded in C++. The output angle became the desired yaw angle of the ASV. The actual yaw of the robot was computed off of the robot's heading, which was measured by the IMU. Proportional control, also known as "P-control," was used to minimize the error, which is the difference between actual yaw and desired yaw, by calculating the efforts of the left and right turning motors, Motors A and B, respectively. The motor efforts were fed into the motor driving code each time the Teensy sampled from the wind vane, and the ASV turned accordingly. The 10Hz sampling rate was appropriate for this system. Samples were only 99 milliseconds apart, so motor efforts were updated about 10 times a second.

Weather vanes point in the direction that wind is blowing from, not where it is blowing to. Although the ASV would be more efficient if it was traveling opposite of where the weather vane was pointing, the decision was made to navigate against the wind to ensure that changes in ASV yaw came from intentional motor efforts instead of the wind blowing it along.

To ensure the wind-navigation system worked, the weather vane and P-control loop were tested in the HMC tank room and at the Scripps College Pool. The ASV was set in the water. The vane angle was suddenly changed, and the ASV's turning response was visually observed.

C. Wind Speed

The minimum wind strength needed to successfully navigate using wind direction can be determined by finding the minimum wind speed that will move the weather vane and push the ASV.

A 3D-printed 3-cup anemometer measured wind speed. The reference design for the anemometer came from the E80 website [10]. The revolutions made by the anemometer per second could be converted to wind speed in miles per hour. A hall effect sensor and a magnet were chosen to determine when a

revolution occurred. The magnet spun at the same rate as the anemometer. It was used to trigger the hall effect sensor and mark one revolution.

The sensor chosen was the AH9246, which is omnipolar and outputs a digital signal [11]. Omnipolar switches will respond to either a strong positive or negative magnetic field. This was a desirable characteristic because any orientation of the magnet would allow the sensor to detect the magnetic field. The digital output also proved useful. The AH9246 outputs 0V when the magnet is near the sensor, and a high voltage, 0.1V less than the supply voltage, when it is not [11]. A drop from the high voltage to no voltage marked one revolution.

The circuit design came from the AH9246 datasheet (Fig. 5) [11]. The supply voltage was 5V, so the high voltage output of the sensor was 4.9V.

At the time of design, it was believed that the digital pins were 5V tolerant. Under that assumption, it was understood that sending 4.9V into a Teensy digital pin would simply output the maximum voltage in the range, 3.3V, and that high and low voltage signals could still be easily read. However, 5V tolerant digital pins were only a feature of the previous Teensy versions [12]. The final design used 5V as the input voltage and operated under the assumption that no shifting or scaling needed to be done before the output voltage of the hall effect sensor was sent to the Teensy pin.

Instrument loading from the Teensy was not expected to interfere with the interpretation of the signal because a high and zero voltage output can easily be differentiated. As a result, a unity gain buffer was not added to this circuit (Fig. 5).

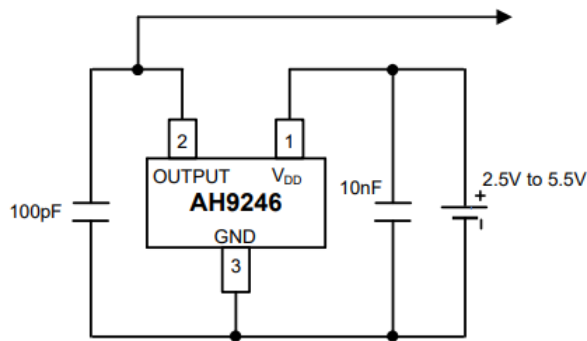


Fig. 5. Hall Effect Sensor Circuit Schematic [11]

The HMC wind tunnel was used to determine a relationship between revolutions per second and miles per hour (mph). Since wind speed cannot be directly set in the wind tunnel, a relationship between fan speed and the speed of wind in the tunnel was determined. Fig. 6 displays the calibration curve and linear equation mapping fan speed to wind speed.

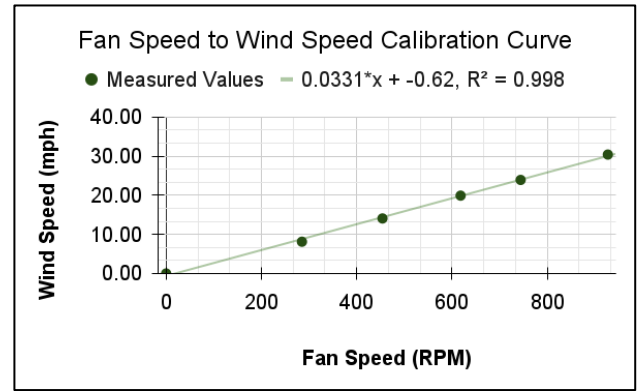


Fig. 6. Wind Tunnel Calibration Curve

At Dana Point, the planned deployment site, in spring 2024, the typical daily range of wind speeds was 0-20 mph [13]. This was the target range for determining the conversion from revolutions per second to wind speed in mph. To test, the anemometer was secured in the wind tunnel test chamber with zip-ties. The chamber was sealed, and the wind was brought up to the desired speed. This step required the use of the equation in Fig. 6. The Teensy recorded the output from the anemometer. After 10-15 seconds of recording at the desired speed, the SD card was removed to stop logging data. This process was performed 7 times with wind speeds ranging from 0-18.6 mph. Wind speeds faster than 18.6 mph were too fast for the anemometer and could have caused damage to the wind tunnel or the sensor itself.

A MATLAB script was written to count the number of revolutions completed by the anemometer. The drop from 3.3V to 0V signified one revolution. Because the Teensy logged data at only one wind speed at a time, the total number of revolutions averaged over the length of the sample in seconds equaled revolutions per second. Wind speed was plotted against revolutions per second, and a linear equation relating the two was produced (Fig. 7).

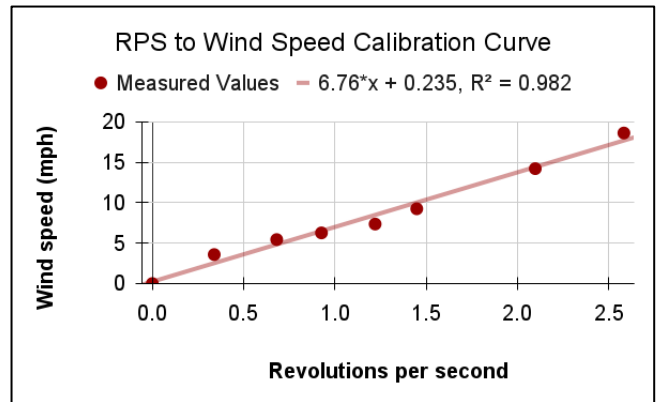


Fig. 7. Anemometer Calibration Curve

The Teensy sampling rate of 10Hz was sufficient for the anemometer. According to Shannon's theorem, the sampling rate should be twice as fast as the highest frequency of the signal. Therefore, speeds faster than five revolutions per second would

not be properly sampled. At the highest wind speed tested, 18.6 mph, the anemometer made about 2.6 revolutions per second.

III. ROBOT DESIGN

A. Mechanical Design

Because the robot is a surface vehicle, it will encounter both waves and wind that can tip it over. Due to these expected conditions, a focus for the ASV body design was stability. The ASV frame has a wide base, and a low center of gravity was created by placing the waterproof box, which contains the motherboard and battery, atop plastic mesh in the center of the PVC frame (Fig. 8). Velcro straps secured the box.

The ASV was also designed with symmetry in mind to keep it balanced. Both the front motors, used to drive the robot forward, and the rear motors, used for turning, were mounted equidistant from the center of the ASV. To ensure the robot was buoyant, pool noodles were placed around each straight segment of PVC pipe (Fig. 8).

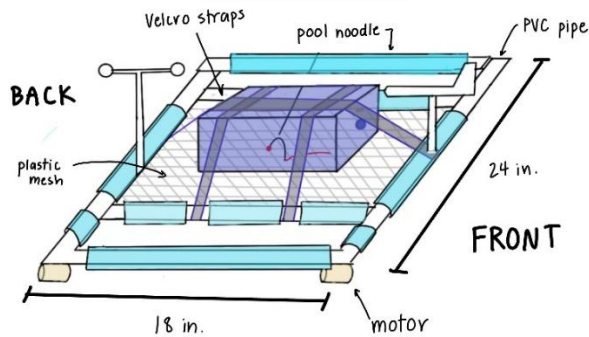


Fig. 8. Mechanical Drawing of ASV. Adapted from [14]

The ASV body also needed to support the wind-measuring devices, which have to be mounted high enough to prevent any interference from water. PVC pipe raised the sensors, and the anemometer was placed higher up than the weather vane to ensure that the two could not collide. To maintain the robot's symmetry, the weather vane was mounted onto the front of the robot in the center, and the anemometer was mounted onto the back middle section (Fig. 9).

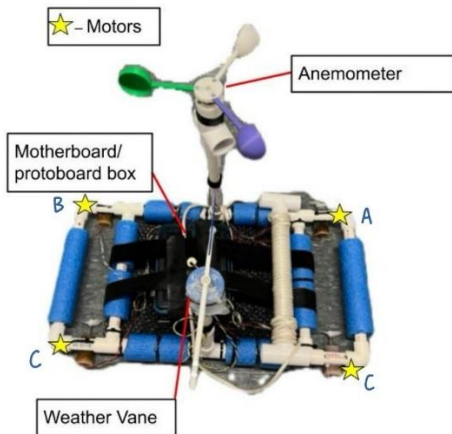


Fig. 9. Placement of Sensors and Motor Labeling (A-C). Adapted from [15]

B. Code Design

The code for the project was mainly written by the E80 course developers [16]. A few modifications were made. The proportional control feedback loop using weather vane output to control motor efforts was added to the existing surface navigation script. All MATLAB code used to process sensor data was original. Also, since vibrations from the motors can cause voltage readings from the Teensy to be noisy, all motors were programmed to turn off for 1 second every 10 seconds. This ensured that noise from the motors could not corrupt all the collected data.

IV. EXPERIMENTAL PROCEDURE

The ASV had a series of tasks to complete during its deployment at Dana Point. The ASV was expected to navigate to a pre-determined starting point using its GPS, stop to collect data, navigate using the wind vane for three minutes, then navigate back to the starting point using GPS.

Before deployment, the position of the robot, in latitude and longitude measurements, was read off of the serial monitor and set as the origin in the code. The desired waypoints of the ASV, which are set relative to the origin of the robot, were selected and hard-coded into the script. The code used to instruct the ASV was uploaded to the Teensy. The weather vane arrow was set at the potentiometer midpoint and was pointed towards the front of the robot. After the position of the weather vane was set, the Teensy was disconnected from the laptop. The waterproof box was sealed and visually inspected to ensure no wires or other components were sticking out of the box. Velcro straps secured the box. The motors were examined and cleared of any debris. The tether was unwound to give some slack before the ASV was placed into the water.

After it appeared that the ASV had completed its course, it was lifted out of the water. The outside of the waterproof box was dried with a towel before opening. The Teensy was then connected to the laptop to determine if the ASV successfully made it to its waypoints and completed its tasks. The SD card was removed from the motherboard, and initial plots of data from the thermistor, anemometer, and vane were observed.

V. RESULTS AND ANALYSIS

A. Deployment Overview

All deployments were conducted from the shore of Baby Beach, in Dana Point, California. To ensure the IMU was functioning properly in the new environment, the magnetometer was recalibrated on-site. The hard and soft iron coefficients were updated in the code.

Morning deployments were done from the beach, and team members waded in the ocean, following the ASV with the tether. In the morning session, debugging the GPS was the focus. The origin, which was set by reading off of the latitude and longitude reported by the GPS, was capable of jumping up to 60 meters away from its original position. The code was edited to set the origin of the ASV as the spot where the GPS initialized, but large jumps in latitude and longitude prevented the ASV from ever reaching its first waypoint. Although the ASV could not demonstrate successful GPS navigation, the anemometer was

able to collect quality data. Wind speed data from one of these trials will be referenced later as Trial 1.

Afternoon deployments were done about 40 meters away from the shore, and the ASV was followed by kayaks, which were rowed by team members wearing life jackets. No laptops or work with electronics was done while sitting in a kayak. During these series of deployments, wind navigation was the focus. GPS navigation was eliminated from the code. The new task for the ASV was to align itself in the direction the wind vane was pointing. The ASV was deployed for periods ranging from 5-15 minutes and collected data from all three sensors. Data from one of these trials, which will be referred to as Trial 2, produced usable data from all three sensors.

To compare H-bridge temperatures between a turning motor and a driving motor, an additional test was conducted. The thermistor was moved from the H-bridge for Motor A, the left rear motor used for turning, to the one on Motor C, a front motor used for forward motion (Fig. 9). The driving motors were turned on, and the ASV was placed in the water. The ASV was held in place with the tether for about 7 minutes. This H-bridge temperature test will be referred to as Trial 3.

B. Thermistor Results

During an afternoon test, Trial 2, which lasted 12 minutes, the temperature of Motor A rose approximately 23.4°C (Fig. 10). The test done for Motor C, Trial 3, which lasted about half the time, resulted in a 36.8°C increase (Fig. 10). The H-bridge of Motor C appeared to heat up at a faster rate than that of Motor A, suggesting that driving motors draw current and consume energy at a faster rate.

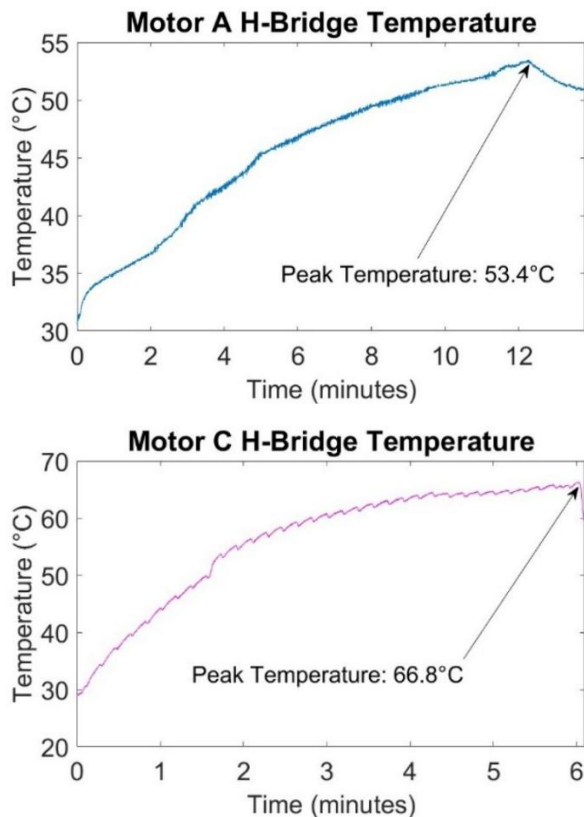


Fig. 10. Motor A (top) and Motor C (bottom) H-bridge Temperatures

To confirm that the thermistor measurements were accurate, ground truth measurements were attempted with a digital thermometer. However, the tip of the probe could not make good contact with the flat H-bridge, resulting in unreliable readings. An infrared thermometer would have been a more appropriate measuring device. To verify that the thermistor circuit was still calibrated with the previously calculated values, the thermistor was dunked into water of known temperatures and compared to expected values.

Although the thermistor data was validated, the results from Trials 2 and 3 cannot confirm that driving motors draw more current than turning motors due to a flaw in the test design. The Motor C terminal was wired with both driving motors during the run. Each motor will draw its own current, and with the data collected, it is not possible to determine how much the H-bridge temperature would have risen with only one motor attached.

C. Anemometer Results

Wind speed data comes from Trials 1 and 2 and was processed using a moving average. Because the wind was not rotating the anemometer multiple times per second, converting revolutions per second to miles per hour at each time step would not yield reliable results. Instead, a moving average with a period of 10 seconds was plotted and served as a more effective way to identify trends in wind speeds.

During Trial 1, the morning run, the average wind speed was 6.35 ± 0.03 mph. In Trial 2, the average wind speed was 6.01 ± 0.02 mph (Fig. 11). Uncertainties were determined using the standard error of the mean.

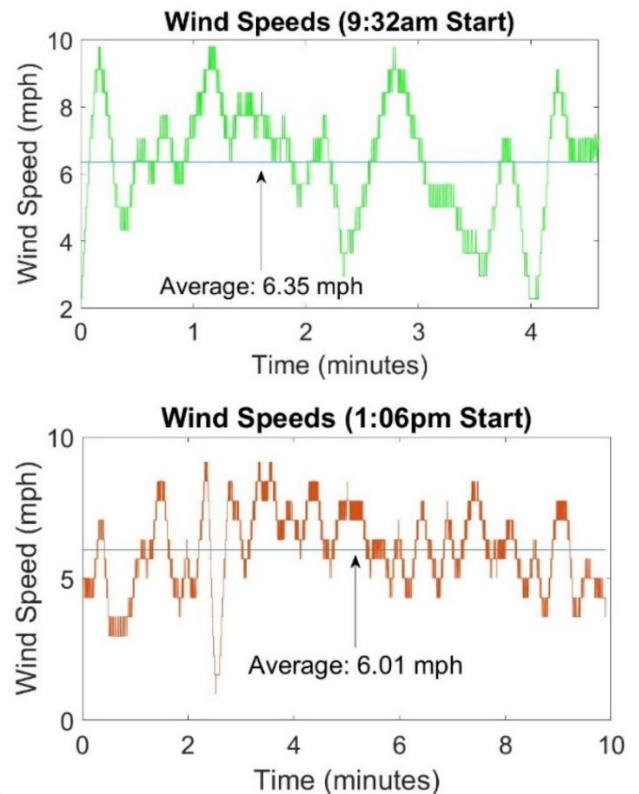


Fig. 11. Morning (top) and Afternoon (bottom) Wind Speed Data

Wind speed data from John Wayne Airport, which is about 25 miles away from Baby Beach, was used to verify results. On the day of deployment, April 26th, around the time of Trial 1, the wind speed was reported to be 8-10 mph [17]. This aligns with the data, as there are several instances where the anemometer is inside the 8-10 mph range (Fig. 11).

During Trial 2, the airport weather station reported wind speeds from 10-12 mph [17]. The measured values do not reach up to this range and stay within 5-8 mph (Fig. 11) [17]. Although the airport data suggests that the wind speeds measured in the afternoon should be higher than those in the morning, these anemometer readings are reasonable. Deployments last for a few minutes and offer only a snapshot of the wind conditions. In addition, wind speeds fluctuate and due to the distance from where the ground truth measurements were taken, the airport data should be treated as an estimate.

During both of these deployments, the wind did not appear strong enough to push the ASV through the water. The wind was strong enough to rotate the weather vane during bigger gusts. However, wind speeds higher than the maximum speed observed, 10mph, would be more ideal for implementing a wind navigation system that takes advantage of the wind to propel the vessel. Wind speeds between 0 and 3 mph do not show up in the data (Fig. 11). The anemometer was too stiff to accurately measure these lower speeds.

D. Weather Vane Results

Trial 2 data was used to assess the performance of the wind-navigation system. An issue with reference frames was discovered when processing the data. The incorrect assumption was made that the weather vane output angle is in the global frame. For example, an output of 0 degrees would indicate the wind is blowing from the North and 90 degrees would indicate West. The vane instead reports wind direction relative to the ASV. Because of the incorrect assumption, the ASV was not navigating itself into the direction that wind is blowing from. Rather, it oriented itself based on the change in angle of the vane. Adding the ASV yaw to the weather vane angle would allow for proper calculation of wind direction in the global frame. Fig. 12 is a plot of wind direction adjusted to be in the correct reference frame. A low-pass filter was applied to reduce noise.

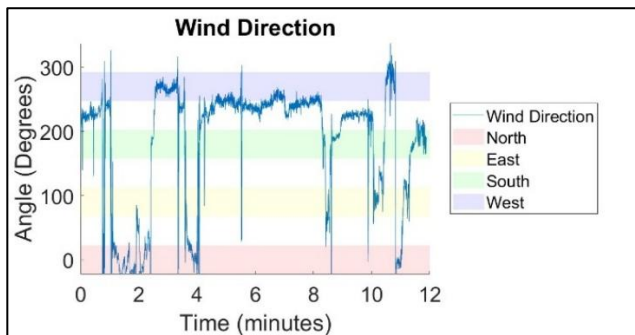


Fig. 12. Wind Direction During Trial 2

The adjusted wind direction data agrees with the John Wayne Airport reports. During the time of deployment, the wind was blowing from the west-southwest and south-southwest

directions [17]. Fig. 12 shows the vane pointing in these directions for approximately 49.3% of the deployment.

Although the ASV was not turning in the direction that wind was blowing, the proportional control code still worked to minimize the difference between desired and actual yaw. During the deployment, it appeared visually that the ASV reacted to changes in wind vane position. A plot of vane angle and actual ASV angle, shown in Fig. 13, also shows a strong correlation between the two angles throughout the trial.

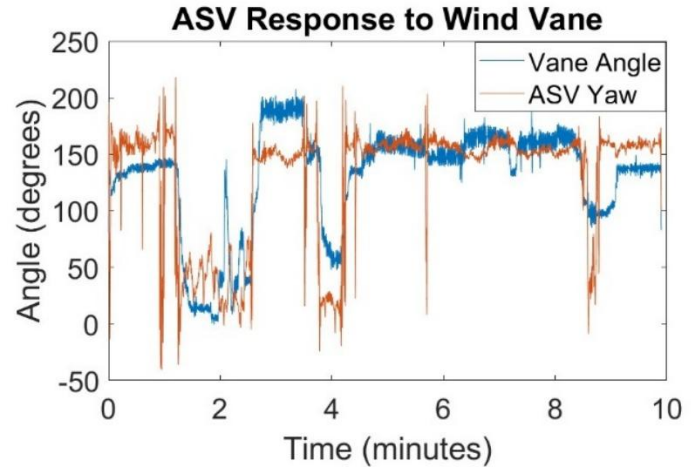


Fig. 13. ASV Yaw Angle Compared to Weather Vane Angle

To more thoroughly assess the accuracy of the control system, an examination into steady state error was performed. Table 1 displays the steady state errors of 6 separate steady-state events during Trial 2. A steady-state event is defined as a period longer than 20 seconds where the desired yaw and actual yaw remain constant. This ensures that the minimum number of samples analyzed is at least 200. The last 15 seconds of the signals in Fig. 14 displays what the beginning of what Event 6 looks like.

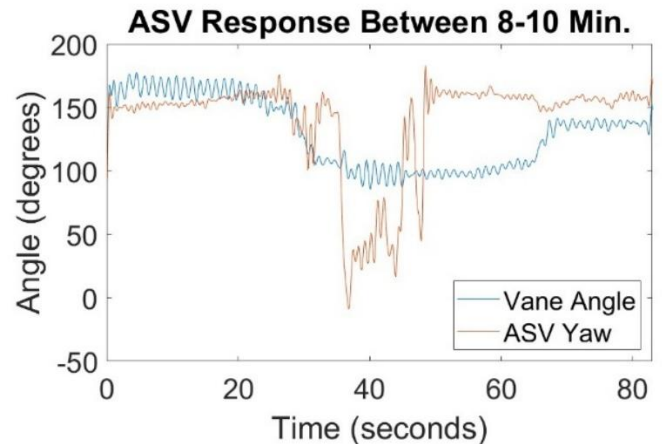


Fig. 14. ASV Yaw Changing in Response to Vane Angle

Using standard error of the mean to find uncertainty, the magnitude of steady-state errors range from 1.2 ± 0.4 to 39.6 ± 0.5 degrees and can occur in both the positive and negative directions (Table 1). Although the upper limit of steady state error is slightly high, there is evidence that motors are adjusting

based on changes in vane angle. In Fig. 14, vane angle changes first, then ASV yaw adjusts accordingly. There is some overshoot, but ASV yaw angle follows changes in vane angle. This finding suggests that a navigation system based on changes in wind direction is feasible.

TABLE I.

Steady-State Error at Six Events During Trial 2		
Event Number	Duration (s)	Steady State Error in Degrees
1	26.4	20.0 ± 0.8
2	44.7	-39.6 ± 0.5
3	49.0	1.2 ± 0.4
4	40.5	9.3 ± 0.9
5	41.2	-7.9 ± 0.4
6	51.7	21.1 ± 0.2

VI. CONCLUSION

A. Summary of Results

Each sensor had its successes and opportunities for improvement.

The thermistor reliably measured temperature, and its calibration curve was verified several times. However, due to the error in Trial 3, the desired data was not collected. The temperature of an H-bridge from one turning motor was to be compared to that of one driving motor. This comparison would have provided insight into which motor draws more current and consumes more energy. Because two driving motors were attached to the H-bridge during the test, a fair comparison of the temperatures was not possible. It is unknown whether turning motors or driving motors are more effective at conserving energy.

The wind navigation system, consisting of the weather vane and proportional control feedback loop, behaved unexpectedly. The ASV did not orient itself in the direction the wind vane pointed, which is the direction that wind blows from in the global frame. The weather vane reported an angle relative to the ASV's frame, and the feedback loop attempted to minimize the difference between this angle and the ASV's yaw angle in the global frame. Despite the reference frame issues, the ASV successfully changed its orientation based on wind vane input, which was the goal of this project. Fig. 13 displays the responses of ASV yaw to changes in weather vane angle. The ASV attempts to match the changes in vane angle throughout Trial 2. Steady-state error went as high as 39.6 ± 0.5 degrees, which is a significant amount of error, but highlights that proportional control can orient the ASV generally in the desired direction.

The wind speed data from the anemometer was generally aligned with information reported by a local weather station. Wind speeds on the higher end of the measured range, those greater than about 8 mph, were capable of rotating the weather vane. Wind speeds higher than what was observed were required to push the ASV through the water. Wind speeds less than 3 mph

were not recorded. It is likely that these winds are not strong enough to rotate the anemometer.

B. Significance of Results

The questions posed about the ASV's feasibility, energy consumption, and operating conditions could not all be thoroughly answered, but results from Dana Point provided key insights.

The overarching goal of the project was to determine if a wind-navigation system was feasible. Although the ASV did not turn in the expected direction, the data gathered at Dana Point demonstrated that the proportional control system was able to turn the ASV in a direction dictated by the wind. The use of low-cost sensors in this build proves that a wind-based navigation system is not only feasible but also accessible.

It was hypothesized that turning motors draw less current and consume less energy than motors that are constantly driving. If turning motors consume more energy in the process of aligning the ASV with the wind than motors that constantly drive in the desired direction, then the claim cannot be made that the wind-navigation system is energy-saving and extends battery life. Since the test done to compare energy consumption was inconclusive due to a flaw in the testing design, claims cannot be made nor refuted.

Operating conditions of the wind-navigation system were also inquired about. An application for this system is a sail-less boat that orients itself in the direction that wind is blowing and turns off its motors if the wind is fast enough to move the boat. Wind speeds were not high enough to push the ASV through the water, so it is unknown what wind conditions are needed for this type of operation. However, wind speeds above 8 mph were able to rotate the wind vane, so the minimum wind speed required to take information about wind direction was discovered.

C. Main Takeaway

Each design decision and assumption made, whether it was for a sensor, the code, or the physical body of the ASV, impacted the deployment and data gathered at Dana Point. The expected ranges of input values to sensors influenced how sensors were developed and calibrated. The code coordinated data collection and ASV movement, and the mechanical design choices ensured the ASV was suitable for deployment in water and that sensors could make measurements without interference. Not all decisions and assumptions were correct or ideal, such as those regarding reference frames, digital pin voltage tolerances, and H-bridge temperature tests. However, the overarching goal of constructing an autonomous surface vehicle that changes its orientation based on input about wind direction was achieved. The use of low-cost sensors in this successful build demonstrates that a wind-based navigation system is possible and does not require advanced technology or inaccessible parts.

D. Future Work

Before the ASV is built on and improved, the tests that were unsuccessful at Dana Point should be redone. The Motor C H-bridge temperature test should be repeated with only one motor connected to the terminal. The proportional control code should be updated to orient the ASV in the direction that wind is

blowing from. Additionally, a 3.3V voltage regulator should be added onto the anemometer circuit to protect the Teensy pins.

If the GPS becomes operational, the ASV could have two navigation modes: GPS and wind. Switching between modes could occur when certain weather conditions are met or when the ASV successfully moves to a waypoint.

Steady-state error could be reduced in future work. This could be done by changing the proportional gain constant or by adjusting motor efforts and placement. Minimizing steady state error would allow for better alignment between the wind vane and ASV yaw angle.

If the proportional control system is refined, wind-navigation could be applied to other types of vessels. The idea of a sail-less boat was previously discussed, but this could also be applied to weather or ocean monitoring probes that study wind patterns.

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CODE APPENDIX: A portion of MainNavigation.cpp adapted from SurfaceControl.cpp [16]. This section converts Teensy output from the potentiometer weather vane to a desired yaw angle. The proportional control loop at the bottom compares the desired angle to the actual yaw angle of the ASV and sets the motor effort variables accordingly.

```
// Convert Teensy reading to voltage

Vteensy = 0.003237 * rawTeensy;

// Get resistance of potentiometer, R1,
from voltage output
Vdivider = -Rn1/Rf*(Vteensy -
(1+Rf/Rn1)* (Rg/(Rpl+Rg)*Vin)); //back
out to get Vout from voltage divider
R1 = R2*(Vin/Vdivider - 1); // back out
to get resistance

// Get R1 in 4000-6000 Ohm range
while (R1 >= 6000 || R1 <= 4000) {
    if (R1 > 6000) {
        R1 -= 1000; // Decrease R1
by 1000 if resistance is greater than 6000
    } else {
        R1 += 1000; // Increase R1
by 1000 if resistance is less than 4000
    }
}
// Get R1 in 0-1000 Ohm range

if(R1>=5000){
    R1 -= 5000;
    angle = R1/1000;
}
else{
    R1 -= 4000;
    angle = R1/1000;
}

// Use the fraction to get vane angle
angledeg = angle*360; // degrees
anglerad = angledeg*3.1415/180;
//radians

// Proportional Control Feedback Loop

yaw = state->yaw;
yaw_des = anglerad;
yaw_error = angleDiff(yaw_des - yaw);
u = Kp*yaw_error;
uR = avgPower + u;
uL = avgPower - u;
```